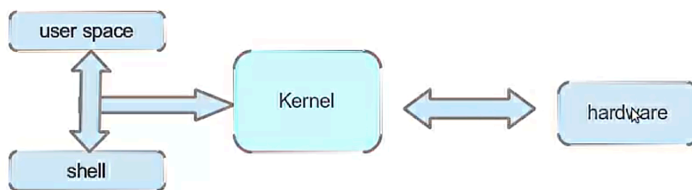


Linux Character Device Driver
Lakshminath Reddy Alamuru
Syracuse University
SUID: 367982169

User Applications, Shell, Kernel, Hardware

Linux Kernel



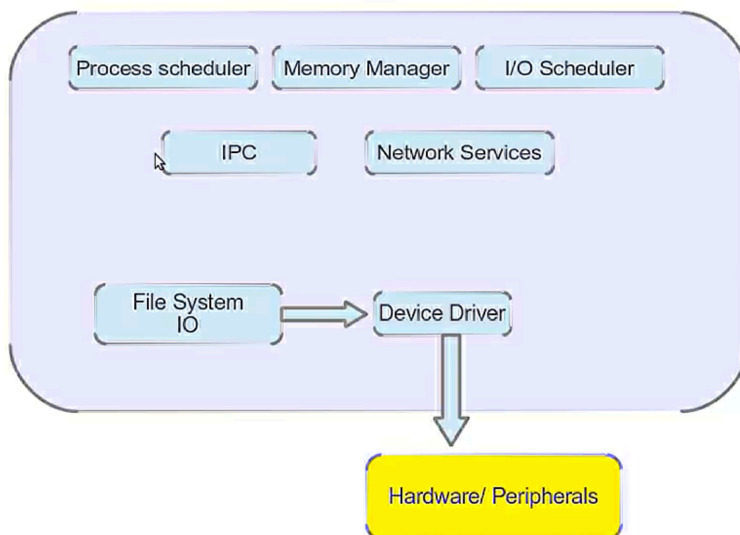
User Applications: Apps

Shell: Invoke program from the shell

Kernel: Process Schedulers, Memory Management, I/O schedulers, IPC, Network Services, Connectivity Services, Filesystem, Device Drivers.

Filesystem <— —> Device Drivers <— —> Hardware Peripherals

Linux Kernel



Linux Kernel Module Programming

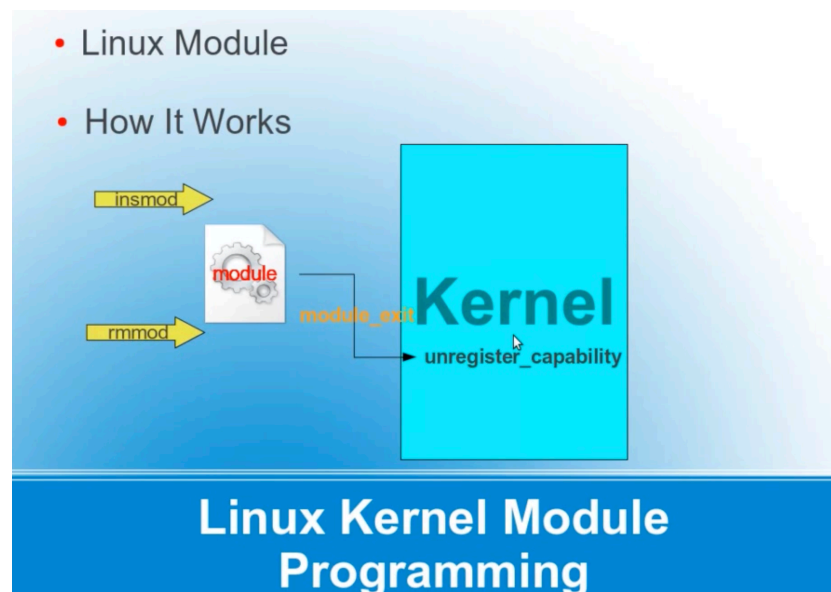
Linux module

Binary code that is inserted into and unloaded out of the static kernel image at run time.

Benefit is to add new functionality on the fly without having to recompile the entire kernel image from scratch and reboot.

Insert Module: `insmod`, `init_module`, `register_capability`

Remove Module: `rmmod`, `module_exit`, `unregister_capability`



Major Number, minor and versions : 3.2.5 are in the order